

SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005

AXALTA COATING SYSTEMS INSTRUMENT SERVICE

Mount Clemens Laboratory
400 Groesbeck Highway
Mount Clemens, MI 48043
Richard Higginbotham Phone: 586 468 9328

CALIBRATION

Valid To: May 31, 2018

Certificate Number: 2111.02

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Chemical

Parameter/Equipment	Range	CMC ² (±)	Comments
pH Meters ³	(4, 7, 10) pH units	0.024 pH units	Reference pH buffers
Conductivity Meters ³	10.00 µS/cm 100.0 µS/cm 1000 µS/cm 10 000 µS/cm	0.36 µS/cm 2.3 µS/cm 6.6 µS/cm 48 µS/cm	Reference conductivity solutions

II. Dimensional

Parameter/Equipment	Range	CMC ² (±)	Comments
Film Thickness Meters ³	(0.4 to 10) mils	5.8 % rdg	Mylar shims
Angle ³ – Level	0° to 180°	0.14°	Electronic level

III. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments
Resistance ³ – Generate	(0 to 999.9) Ω (1 to 999.9) kΩ (1 to 1.9999) MΩ (2 to 9.99) MΩ (10 to 19.99) MΩ (20 to 99.9) MΩ	0.12 % rdg + 0.04 Ω 0.12 % rdg 0.12 % rdg 0.084 % rdg + 4.5 kΩ 0.24 % rdg 0.16 % rdg + 120 kΩ	Decade box
DC Voltage ³ – Measure	(0 to 199.99) mV (0.2 to 1.9999) V (2 to 19.999) V (20 to 199.99) V (200 to 999.9) V	0.04 % rdg + 0.022 mV 0.04 % rdg + 0.22 mV 0.05 % rdg + 2.3 mV 0.05 % rdg + 23 mV 0.05 % rdg + 0.25 V	Fluke 8060A DMM
Electrical Calibration of Thermocouple Devices ³ – Type J Type K Type T	 (-210 to -100) °C (-100 to 800) °C (800 to 1200) °C (-200 to -100) °C (-100 to 400) °C (400 to 1200) °C (1200 to 1372) °C (-250 to -200) °C (-200 to 0) °C (0 to 400) °C	 0.53 °C 0.42 °C 0.53 °C 0.61 °C 0.52 °C 0.52 °C 0.52 °C 1.1 °C 0.62 °C 0.52 °C	 Fluke 754 process calibrator
Electrical Calibration of RTD Indicating Systems ³ – Pt 100	 (-200 to 100) °C (100 to 800) °C	 0.04 °C 0.01 % rdg + 0.03 °C	 Fluke 754 process calibrator

IV. Fluid Quantities

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments
Viscometer ³ –			
Rotational Devices	(20 to 99) % of Full Scale	4.1 cP	Cannon reference oil
Cone & Plate	(80 to 900) cP	0.7 % rdg + 0.14 cP	
Krebs Unit	(61 to 106) KU	0.22 % rdg + 0.05 KU	
Viscosity Cup Devices ³	18 mm ² /S (cST) 34 mm ² /S (cST) 65 mm ² /S (cST) 120 mm ² /S (cST) 227 mm ² /S (cST) 395 mm ² /S (cST) 722 mm ² /S (cST)	0.7 s 0.6 s 0.5 s 0.5 s 0.5 s 0.7 s 1.0 s	Cannon standard oils, reference stopwatch, temperature controlled bath

V. Mechanical

Parameter/Equipment	Range	CMC ² (±)	Comments
Balances ³ –			
4 Decimal	(1 to 200) g	1.7 µg/g + 110 µg	Ultra Class weights Class 1 weights Class 1 weights Class 1 weights Class 1 weights Class 1 & 2 weights
3 Decimal	(1 to 1200) g	1.8 µg/g + 1.3 mg	
2 Decimal	(1 to 8000) g	1.5 µg/g + 13 mg	
1 Decimal	(1 to 30 000) g	0.72 µg/g + 58 mg	
0 Decimal	(1 to 30 000) g	0.6 g	
	(30 000 to 60 000) g	1.1 g	

Parameter/Equipment	Range	CMC ² (±)	Comments
Pendulum Hardness ³ – Testers			
Konig Hardness	(98 to 102) Oscillations (171 to 186) Oscillations	72 ms 2.1 s	Reference stopwatch & BYK standard glass plate
Persoz Hardness	(99 to 101) Oscillations (415 to 445) Oscillations	0.13 s 1.4 s	
Pressure ³ – Measure (Pneumatic)	(0 to 60) psig (60 to 300) psig	0.033 psig 0.051 % rdg + 0.02 psig	Digital pressure gauge
Arizona Instruments Moisture Analyzers ³ –			
Temperature	(25 to 100) °C (100 to 275) °C	1.0 °C 0.3 % rdg + 0.7 °C	Process calibrator w/T TC special limits
Balance	(1 to 20) g	5.6 µg/g + 150 µg	Class 1 weights

VI. Optical Quantities

Parameter/Equipment	Range	CMC ² (±)	Comments
DOI Meters ³ – Hunter, Tricor	(80.0 to 100.0) Rs	2.2 Rs	Tricor supplied Rs standard
Gloss Meters ³ –			
20 degrees	(4.0 to 100.0) GU	1.4 GU	BYK calibrated gloss tiles
60 degrees	(4.0 to 100.0) GU	1.4 GU	
85 degrees	(4.0 to 100.0) GU	1.4 GU	

Parameter/Equipment	Range	CMC ² (±)	Comments
Wavescan Meters ³	(0.0 to 63.0) Wa (0.0 to 59) Wb (0.0 to 44.0) Wc (0.0 to 49.0) Wd (26.0 to 62) We (0.0 to 40.0) Lw (0.0 to 50) Sw (2.0 to 46.0) du	2.1 Wa 2.4 Wb 2.4 Wc 2.6 Wd 2.1 We 2.0 Lw 2.3 Sw 2.3 du	BYK reference tiles
Correlated Color Temperature ³ – Measure	(2000 to 8000) K	2.1 % rdg	X-Rite i1Pro Rev. E
Turbidity Meters ³	(1 to 4000) NTU	5.9 % rdg	StablCal Formazin standard solutions
Illuminance ³ – Measure	(500 to 5000) lx	1.8 % rdg	X-Rite i1Pro Rev. E

VII. Thermodynamics

Parameter/Equipment	Range	CMC ² (±)	Comments
Temperature ³ – Measuring Equipment	(25 to 300) °C	(0.25 % rdg + 0.19) °C	Heat block, process calibrator w/RTD
Temperature ³ – Measure	(-50 to 0) °C (0 to 70) °C (70 to 100) °C (100 to 400) °C (400 to 900) °C	0.29 °C 0.10 °C 0.21 °C 0.23 % rdg + 0.18 °C 0.87 % rdg + 0.07 °C	Process calibrator Digital thermometer Process calibrator w/RTD or w/TC
Relative Humidity – Measure	(10 to 80) % RH	2.1 % RH	Reference thermohygrometer

VIII. Time & Frequency

Parameter/Equipment	Range	CMC ² (±)	Comments
Timers/Stopwatches ³	(5 to 7200) s	0.0003 % rdg + 47 ms	Direct comparison reference stopwatch

¹ This laboratory offers commercial calibration service and field calibration service.

² Calibration and Measurement Capability (CMC) uncertainty is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC uncertainty due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ Field calibration service is available for this calibration and this laboratory meets A2LA R104 – General Requirements: Accreditation of Field Testing and Field Calibration Laboratories for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC uncertainty found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC uncertainty.

⁴ The stated measured values are determined using the indicated instrument (see Comments). This capability is suitable for the calibration of the devices intended to measure or generate the measured value in the ranges indicated. CMC's are expressed as either a specific value that covers the full range or as a percent or fraction of the reading plus a fixed floor specification.



Accredited Laboratory

A2LA has accredited

AXALTA COATING SYSTEMS INSTRUMENT SERVICE

Mount Clemens, MI

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 19th day of May 2016.

A handwritten signature in blue ink, appearing to read 'J. C. Bunt', is shown above a horizontal line.

Senior Director of Quality and Communications
For the Accreditation Council
Certificate Number 2111.02
Valid to May 31, 2018

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.